

1 Key Features

Functional:

- Counter anode pulse registration
- Universal connection via I2C
- Support for two radiation intensity calculation algorithms
- Dynamic counting time period adjustment
- Total pulse count measurement
- Software change of I2C address
- Standalone use as a radiation indicator
- Logic level 3.3V. Tolerant to 5V level (can be used without a level converter)
- Compatible with any counter with operating voltage ~400V

Electrical:

- Low supply voltage 3.0...3.5 V
- Maximum current consumption at high radiation - no more than 100 uA

Technical:

- Compact module dimensions 89mm x 21mm x 13.5mm
- Fixed (vibration-resistant) counter placement
- Module weight no more than 12 g
- Operating temperature range from -20°C to +60°C

2 Description

RadSens is a universal dosimeter-radiometer in a modular form factor. A gas-discharge or mica Geiger-Muller counter can be used as the sensing element in the module.

The device supports measurement and calculation of radiation intensity using two algorithms: with a dynamic counting time range for detecting local contamination sources, and with a wide static time range for accurate measurement of the current background radiation level. It is also possible to use the module without additional devices as a radiation "indicator" by observing the blinking frequency of the on-board LED.

Pulse registration, calculation algorithms, and data transfer via I2C with bus operating frequency up to 100 kHz are implemented on the on-board STM32 microcontroller. The module supports software address change. It is also possible to adjust the counter sensitivity to ionizing radiation via I2C.

At startup, the dosimeter blinks the LED, indicating that it has entered operating mode.

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3 Device Specifications

3.1 Technical

Overall device dimensions with counter installed: 89mm x 21mm x 13.5mm for model RSb and 120 x 21 x 20 for version RSb-h. Module weight: 12 grams.

Parameter	Value			Unit
	min	nominal	max	
Supply voltage	3,0	3,3	3,5	V
Maximum current consumption	-	80	100	uA
Anode voltage on gas-discharge counter	380	400	440	V
Operating temperature range	-40	+20	+70	°C
Operating humidity range	0	60	98	%

Table 1 (Technical Specifications)

3.2 Metrological

By default, the module sensitivity is configured for the more common Geiger counters SBM20-1 or SBM20 manufactured by SF JSC "NIITFA", decimal number TDMK.433217.008, compliant with technical specifications OD0.339.544TU. Radiation

intensity is calculated using the formula: $RAD = N \times \frac{60_{min} \times 60_{sec}}{P_{cp} \times dT}$, where

P_{avg} is the average sensitivity of the SBM20-1 (SBM20) counter to gamma radiation from a Ra^{226} ,

dT is the time interval for registering the number of pulses,

N is the number of pulses registered during time dT ,

RAD is the radiation activity value in uR/h.

Parameter	Value			Unit
	min	nominal	max	
Measurable radiation range	14,4	-	144 000,0	uR/h
Number of pulses between data reads	0	-	65 535	pulses
Sensitivity to gamma radiation Ra^{226}	100	105	110	pulses/uR
Relative sensitivity spread	-	-	±15	%

Table 2 (Metrological Specifications)

4 Data Communication

4.1 Register Map

Data exchange (configuration and transmission of measured values) is performed via the I2C interface at speeds up to 100 kHz. The sensor operates in Slave c mode with default address 0x66 (software configurable). Logic level is 3.3V. Tolerant to 5V level (can be used with 5V devices without a level converter).

Address	Name	R/W	Range	Unit
0x00	Device ID	R	0x7D	-
0x01	Firmware version	R	0-255	-
0x02	<reserved>	-	-	-
0x03-0x05	Radiation intensity (measurement period T < 123 csec.)	R	0 ... 1 440 000	0,1*uR/h
0x06-0x08	Radiation intensity (measurement period T = 220 csec.)	R	0 ... 1 440 000	0,1*uR/h
0x09-0x0A	Pulse counter (reset on read)	R	0 ... 65535	pulses
0x0B-0x0F	<reserved>	-	-	-
0x10	Address Device	W	0x03-0x77	-
0x12-0x13	Counter sensitivity	R/W	0-65535	pulses/uR
0x14	Indicator LED control	R/W	0/1	-

Table 3 (Data Communication Register Map)

4.2 Register Description

4.2.1 ID Device

[address: 0x00, size: 8 bits, access: R]

Control register containing the device identifier. Default value is 0x7D. Used to verify device connection.

4.2.2 Firmware version

[address: 0x01, size: 8 bits, access: R]

Register storing the current firmware version. Used for monitoring and timely software updates.

4.2.3 Radiation Intensity (Dynamic Counting Period)

[address: 0x03, size: 24 bits, access: R]

Contains the dynamic value of ionizing gamma radiation intensity. When a sharp change in radiation intensity is detected (either increase or decrease), the counting period of the sliding window is dynamically adjusted so that the range covers only the time interval containing current data. Allows using the device in local contamination search mode. Update rate: 1 sec.

4.2.4 Radiation Intensity (Static Counting Period)

[address: 0x06, size: 24 bits, access: R]

Contains the statistical value of ionizing gamma radiation intensity. The sliding window counting period is 220 sec. Allows accurate measurement of constant background radiation. Update rate: 1 sec.

4.2.5 Pulse counter

[address: 0x09, size: 16 bits, access: R]

Contains the accumulated number of pulses registered by the module since the last data read via I2C. The value is reset each time it is read. Allows direct processing of Geiger counter pulses and implementation of other algorithms. The value is updated at the moment each pulse is registered.

4.2.6 Address Device

[address: 0x10, size: 8 bits, access: W]

This register is used to change the device address when multiple devices need to be connected to the same bus. Default value is 0x66. After writing, the new value is saved to the microcontroller non-volatile memory.

4.2.7 Counter sensitivity

[address: 0x12, size: 16 bits, access: R/W]

Contains the Pavg coefficient value (section 3.2) used to calculate radiation intensity. If needed (e.g., when installing a different type of counter), the required sensitivity value in pulses/uR is written to the register. The default value is 105 pulses/uR. After writing, the new value is saved to the microcontroller non-volatile memory.

4.2.8 Indicator LED control

[address: 0x14, size: 8 bits, access: R/W]

Indicator LED control register (LED is located on the module board). Enabled by default. To enable indication, write 1 to the register; to disable, write 0. If other values are written, the command is ignored.

4.3 Pulse Output

4.3.1 Description

Designed for pulse registration by an external device (controller).

4.3.2 Operating Procedure

Line operating level is 3.3 V. When a pulse is registered, the module pulls the line to 0 for 11.2 ms, then restores the high operating level. Delay from counter pulse registration to signal generation on the line is 15 us.

5 Connection Connector

The board has an "XH-4A" connector installed, mating part: "xh2.54-4p". The connector pinout is shown in the table below. The separate INT pin is not listed in the pinout. The operation of this output is described in section 4.3.

Pin	Name	Function
1	VCC	Sensor power supply
2	GND	Common ground
3	I2C-SCL	I2C interface clock line I2C
4	I2C-SDA	I2C interface data line I2C

Table 4 (Connection Connector Pinout)

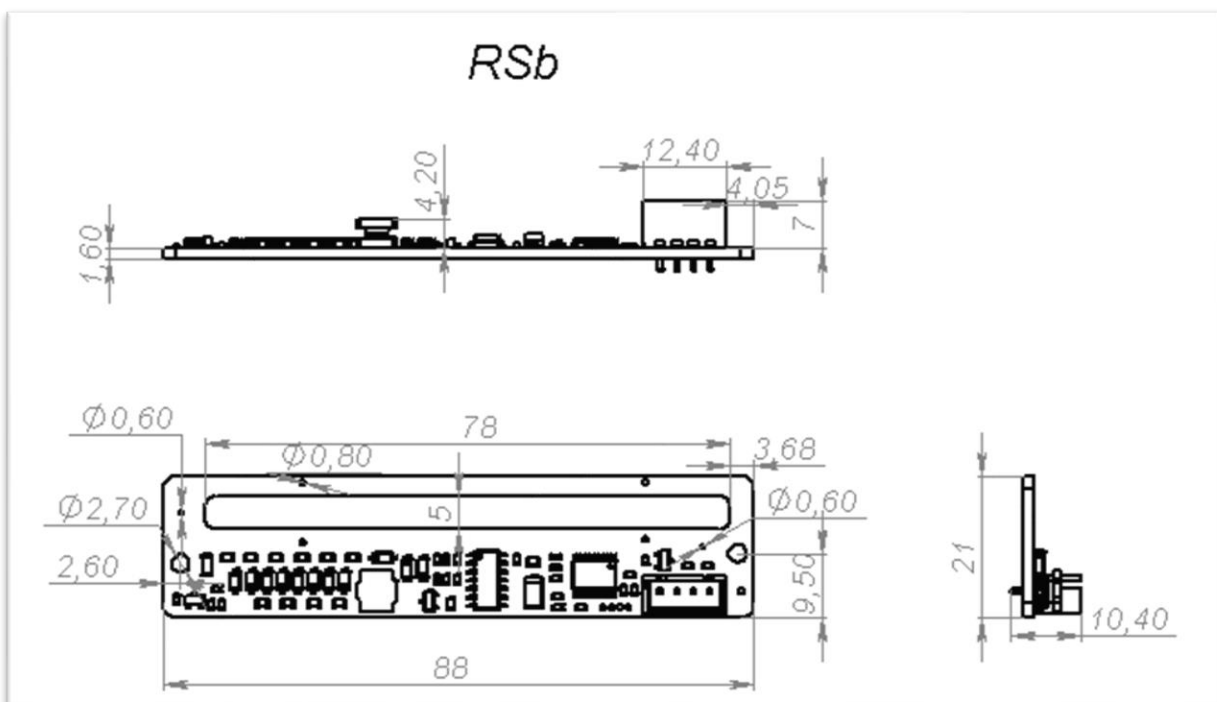
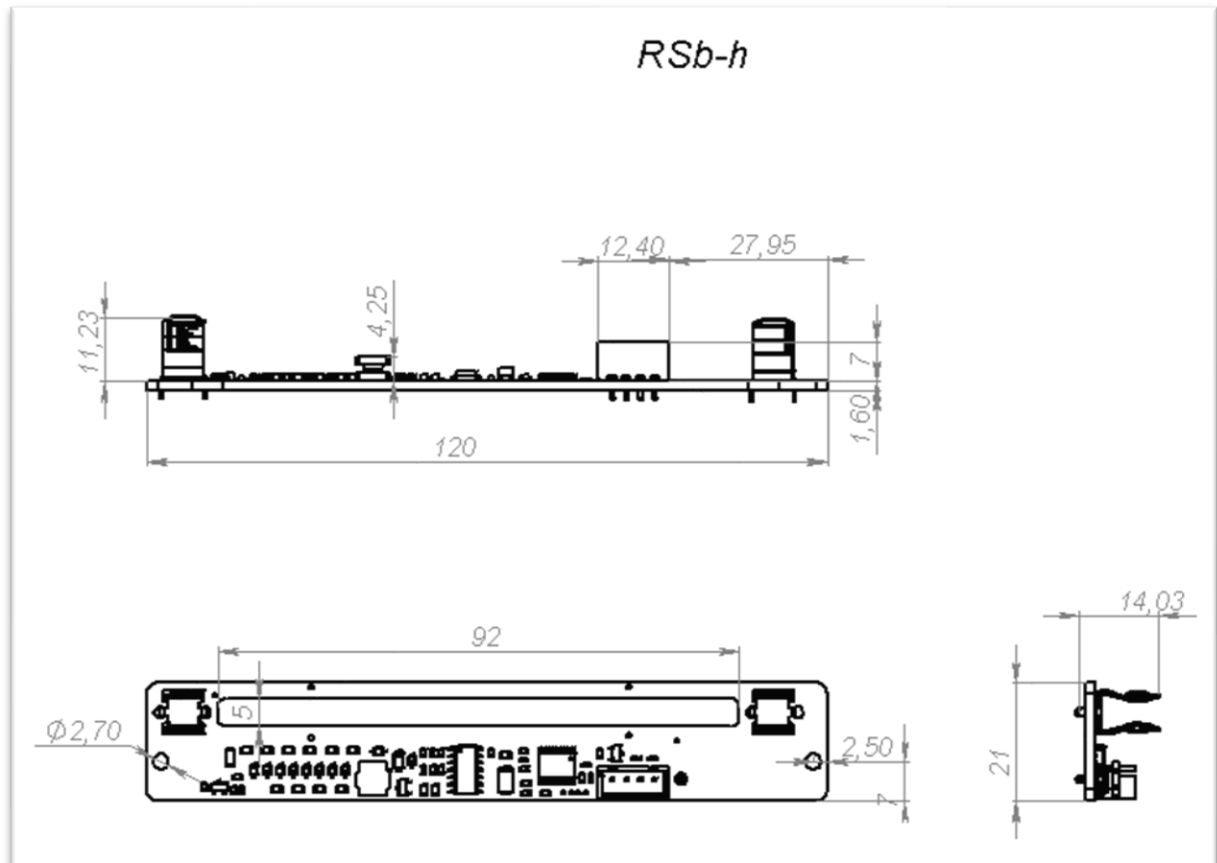
6 Product Range

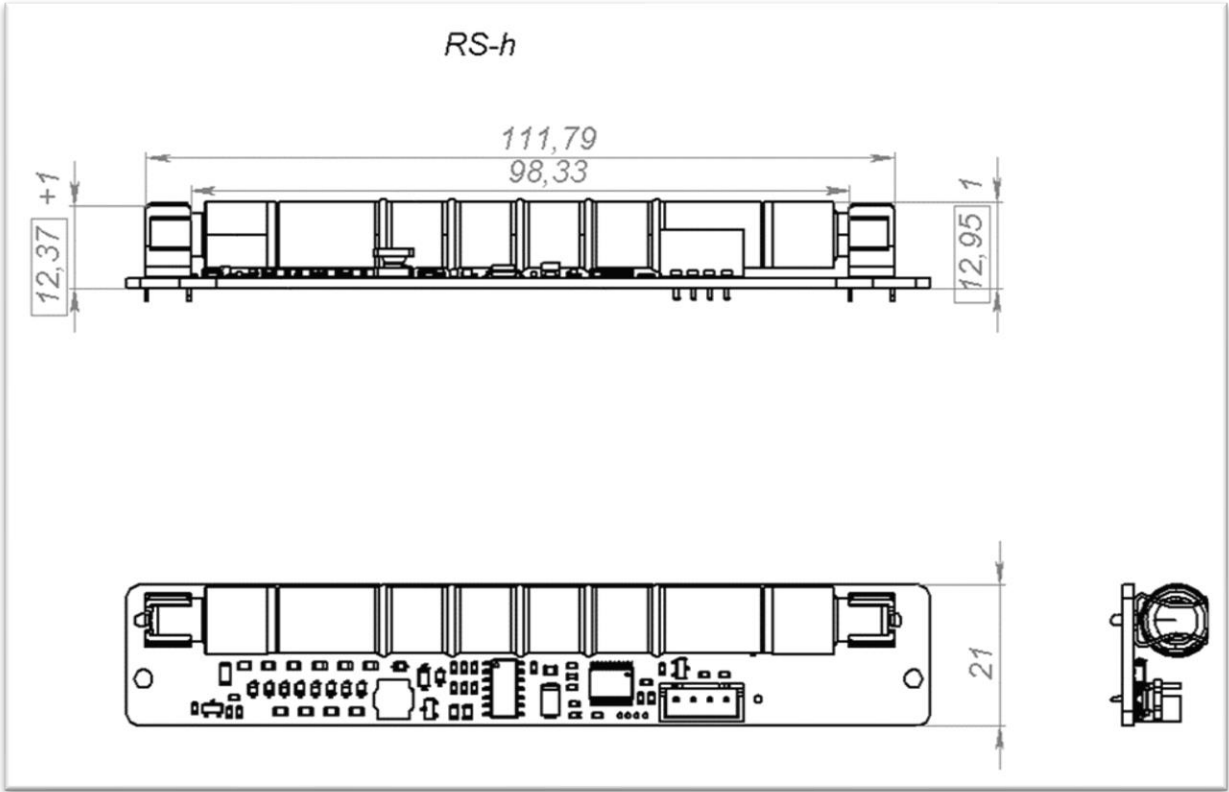
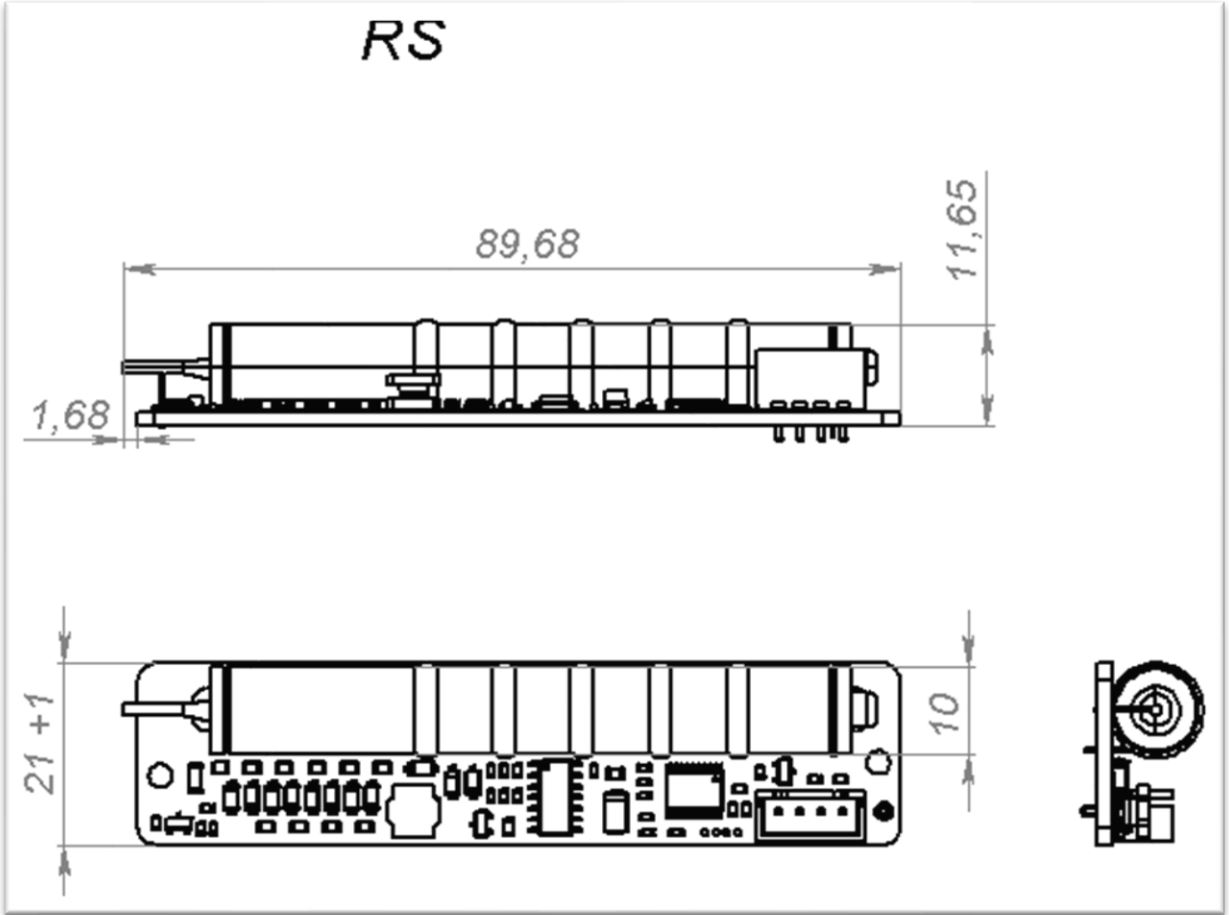
The dosimeter is manufactured in 4 versions. Version descriptions are provided in the table below.

Part number/ Article number	Description
RS	Compact module version with SBM20-1 counter installed. Counter connection method: soldering.
RSb	Compact module version, counter not included. Counter installation by soldering
RSh	Enlarged module version, supplied with SBM20 counter. Counter is secured in special quick-release holders.
RSbh	Enlarged module version, without counter installed. Has 2 special holders for gas-discharge tubes, as well as 2 soldering holes.

Table 5 (RadSens)

7 Module Drawing





8 Additional Resources

Contact information and module operation details are provided in the table below.

Description	Link
Manufacturer website	http://climateguard.ru/
Module library	https://github.com/climateguard/RadSens

Table 6 (Useful Resources)